

Algebra II - Tutorial 6 - Date: 2/3/2026

1. How many elements are there in $G = SL_2(\mathbb{Z}_5)$. is there a subgroup of G such that $x^4 = 1$, $y^5 = 1$ and $xy = y^2x$.
2. Describe the class equation of all groups of order 12.
3. Show that a group of order $2p$ where $p > 2$ is prime is dihedral or cyclic group.
4. Describe all groups of order 55.
5. Let H be a proper subgroup of G . Prove:
 - (a) The group G is not the union of all conjugate subgroups of H .
 - (b) There is a conjugacy class disjoint from H .
 - (c) Let P denote the subgroup of $GL_n(\mathbb{R})$ consisting of permutation matrices. Determine the normalizer $N(P)$.
6. Let G be a p -group. Show that the normalizer of H , $N(H) := \{g \in G : gHg^{-1} = H\}$ is strictly larger than H and that H is contained in a normal subgroup of index p .
7. Compute the order of $GL_n(F_p)$ and determine the number of sylow- p subgroups.
8. Let G be a group of order 28.
 - (a) Determine the number of sylow-2 subgroups and sylow 7 subgroups.
 - (b) Suppose G is non-abelian, and the sylow-2-subgroup is cyclic.
 - i. Determine the number of sylow 2 subgroups and the number of sylow-7 subgroups.
 - ii. Prove that there is at most one isomorphism class of such groups.
 - iii. Determine the number of elements of each order and the class equation of G .